

Arvid Sandberg

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EDUCATION

University of Waterloo | *B.A.Sc. in Systems Design Engineering with specialization in Financial Technology* September 2017 - May 2021

- Completed coursework in numerical methods, computational finance, and stochastic calculus; GPA 3.78/4.0 with Dean's List recognition in 6 of 8 semesters.

EXPERIENCE

Citadel Securities | *Quantitative Developer* June 2022 - Present

- Engineered real-time market microstructure analysis system processing 50M+ tick events daily, reducing latency from 2.3ms to 340µs through vectorized NumPy operations and Numba JIT compilation.
- Developed Monte Carlo simulation framework for pricing exotic derivatives, achieving 15x speedup by implementing GPU-accelerated matrix operations with CuPy across 1,024 parallel paths.
- Optimized portfolio risk model by refactoring Cython modules and implementing adaptive grid refinement, cutting computation time from 45 minutes to 3.2 minutes for daily VaR calculations across 10,000+ positions.

Jane Street | *Quantitative Developer (Mid-Level)* August 2021 - May 2022

- Built statistical arbitrage backtesting engine in OCaml with 18-month historical dataset, identifying 22 new profitable strategies with Sharpe ratios exceeding 1.8 across four asset classes.
- Debugged and optimized core price discovery algorithm, resolving race conditions in multiprocessing pipeline that improved tick-to-order fill consistency by 12% under high-volume market conditions.
- Collaborated with traders to validate 6 machine learning models for regime detection, implementing feature engineering pipeline that improved model accuracy from 68% to 81% on out-of-sample data.

Numerai (Quantitative Research Division) | *Junior Quantitative Developer* January 2021 - July 2021

- Constructed ensemble meta-model combining 40+ individual predictors using LightGBM stacking, achieving 15th percentile ranking on Numerai tournament with monthly payout of \$2,400 USD.
- Implemented automated data validation pipeline detecting missing values and outliers in 2.1M+ historical stock features, reducing downstream model training failures by 34%.
- Accelerated correlation matrix computations from 22 seconds to 1.8 seconds by migrating from pandas to polars and vectorizing covariance calculations for 8,000-stock universe.

PROJECTS

Reinforcement Learning Market Maker | *Python, TensorFlow, Ray, PostgreSQL* October 2023 - March 2024

- Developed DQN-based market making agent using historical LOB data from 3 years of E-mini S&P 500 futures, achieving net P&L of \$47,300 in live paper trading on 50-contract portfolio.
- Engineered feature extraction pipeline normalizing 18 microstructure indicators (spread, depth skew, order flow imbalance) across multiple asset classes with 99.2% data integrity validation.

High-Frequency Volatility Surface Interpolator | *C++, Eigen, HDF5, SciPy* May 2023 - August 2023

- Constructed radial basis function (RBF) interpolation system for 2D equity option volatility surface supporting 12,000+ data points, reducing pricing errors to 1.3 basis points vs. 28bp with linear methods.
- Optimized sparse matrix operations reducing memory footprint from 2.8GB to 340MB, enabling real-time surface recalibration every 2 seconds across 50+ underlyings.

Cross-Exchange Statistical Arbitrage Framework | *Python, Pandas, Scikit-learn, Redis* January 2023 - April 2023

- Built low-latency cointegration detection engine monitoring price deviations across 8 cryptocurrency exchanges, identifying 14 profitable pairs with average holding period of 8 minutes and 62% win rate.
- Implemented vector autoregression model with rolling Kalman filter to dynamically adjust hedge ratios, improving P&L consistency and reducing maximum drawdown from 18% to 9.2%.

EXTRACURRICULAR

Global Association of Risk Professionals (GARP) | *FRM Charterholder* March 2023 - Present

- Completed Financial Risk Manager certification covering market, credit, and operational risk; scored in 88th percentile on Part II exam.
- Mentoring 3 junior developers through FRM study materials, focusing on quantitative risk concepts and practical applications in derivatives pricing.

QuantConnect Community | *Algorithm Author & Technical Reviewer* September 2022 - Present

- Published 5 open-source trading algorithms achieving average 1,200+ views and 156 GitHub stars across community, with 3 algorithms featured in QuantConnect Academy tutorials.

- Reviewed 28+ peer submissions improving code quality, backtesting methodology, and documentation standards across C#, Python, and F# implementations.

TECHNICAL SKILLS

Languages – English (fluent), Swedish (native), German (intermediate)

Technologies – Python (NumPy, Pandas, SciPy), C++, Cython, OCaml, TensorFlow, PyTorch, Numba, SQL (PostgreSQL, ClickHouse), Redis

Tools – Git, Docker, Kubernetes, Jupyter, CMake, gdb, perf, Bloomberg Terminal, FactSet, Refinitiv